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A

B.Tech. Degree V Semester Supplementary Examination April 2018

CS 15-1504 OPERATING SYSTEM

(2015 Scheme)

Time : 3 Hours

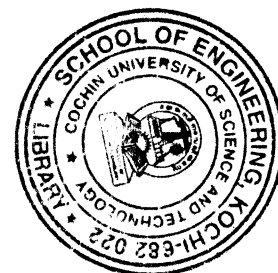
Maximum Marks : 60

PART A

(Answer **ALL** questions)

(10 × 2 = 20)

- I. (a) What is a system call? Give an example.
- (b) What is a process? With a neat diagram write about various states of a process.
- (c) What is meant by guaranteed scheduling? Explain.
- (d) What is meant by inverted page table? Compare it with normal page table.
- (e) Differentiate between paging and segmentation.
- (f) What is meant by deadlock? What are the necessary conditions for a deadlock to occur?
- (g) What is meant by non resource deadlock? Explain with an example.
- (h) Explain briefly interrupt handlers.
- (i) Write briefly about file system implementation in Linux operating system.
- (j) Discuss about the applications of real time systems.



PART B

(4 × 10 = 40)

- II. (a) Explain inter process communication. (5)
- (b) What is critical section problem? Explain the use of semaphores in solving the producer consumer problem. (5)
- OR**
- III. (a) Explain with an example any two CPU scheduling algorithms. (5)
- (b) Write notes on (i) Context switch and (ii) two level scheduling. (5)
- IV. (a) Explain the following terms related to memory management. (5)
- (i) Swapping.
- (ii) Compaction.
- (b) Explain about any two memory management mechanisms. (5)
- OR**
- V. (a) Explain with an example how address mapping is done in paging. (5)
- (b) What is the need for page replacement? Explain any two page replacement algorithms with an example. (5)
- VI. (a) Explain various deadlock prevention mechanisms. (5)
- (b) Explain deadlock detection in a system with resources having single instance. (5)
- OR**
- VII. Explain with an example the Banker's algorithm for deadlock avoidance in a system with resources having multiple instances. (10)
- VIII. (a) Explain with an example any two disk scheduling algorithms. (5)
- (b) Explain briefly I/O buffering. (5)
- OR**
- IX. (a) Explain scheduling in RTOS. (5)
- (b) Explain the characteristics of a real time operating system. (5)